

CSCE 671 HUMAN-COMPUTER INTERACTION · MODULE 11 · LECTURE NOTES

Input and Output

Modalities, and what each one is good for.

MODULE OUTCOMES

- Explain Fitts's law and what it predicts.
- Explain the input modalities and their error characteristics.
- Explain latency's effect on interaction.
- Explain modality appropriateness.
- Choose an interaction for a device and a context.

1 Pointing

$$T = a + b \cdot \log_2(D/W + 1)$$

- **The time to acquire a target**, from the distance to it and its width along the direction of travel — and it is **the one genuinely quantitative and predictive law in interface design**, which makes it worth knowing precisely.
- **So bigger is faster and nearer is faster, logarithmically in both** — which means **doubling a target's size helps less than halving the distance to a small one**, and the logarithm is why very large targets stop helping.
- **And an edge target has effectively infinite width** along the direction of travel, **because the pointer cannot overshoot it** — which is why a **screen-edge menu bar is measurably faster to hit than a floating one of the same visible size**.
- **The edge consequence is the non-obvious one**, and it is the most useful: **a target you cannot overshoot behaves as though it were much larger**, which is a free improvement available to any design that can place something at an edge or a corner.
- **Make frequently-used targets larger**, and **dangerous ones smaller or further away** — which is **Module 03 §4's slip prevention with a number attached** and is a quantitative design decision rather than a judgement.
- **Put things at the edges and the corners**, which are effectively infinite targets — and **corners are the fastest points on a screen**, being infinite in two directions at once.
- **Keep related controls near where the pointer already is**, which is **why a context menu beats a toolbar for a selection-specific action**: the distance is zero by construction.

- **And beware of adjacent targets**, since **the law predicts the travel time and says nothing about the accuracy — adjacency is what produces slips** (Module 03 §4), so speed and safety pull apart here and have to be traded deliberately.
- **On touch, the minimum target size is set by the finger** rather than by the display — roughly nine millimetres regardless of resolution — **which is much larger than a cursor needs** and is why a dense desktop layout cannot simply be scaled. **Dangerous targets should be smaller and further away**, which is **the deliberate use of the law against itself** and is the right design for a destructive action.

2 Modalities

Modality	What it is good for	Its error profile
Keyboard	Text entry, and fast expert command invocation.	Typos; and the commands are discoverable only if signified (Module 03 §3).
Mouse or trackpad	Precise pointing, selection, and dragging.	Slips on adjacent targets (§1).
Touch	Direct manipulation, and anything mobile.	Occlusion by the hand, imprecision, and no hover state at all — see the note.
Speech	Hands-free and eyes-free operation.	Recognition errors, and no undo within the channel (see the callout).
Gaze	Selection when no other channel is available at all.	Involuntary movement, and no natural equivalent of a click.

'No hover' is the touch constraint most often forgotten — every tooltip, every preview, every hover-to-reveal control, and every disabled-state explanation that depended on hovering is simply unavailable, and the no-hover consequence breaks a great many desktop design patterns that are otherwise portable. The usual response is to move the information into a tap, which costs a step; the better response is to make it unnecessary.

SPEECH HAS NO CHEAP UNDO, WHICH CHANGES THE WHOLE INTERACTION

A misrecognised word must be corrected by speaking again — which is slower than the original input was and may itself be misrecognised, so errors compound rather than resolve, which is a qualitatively different failure mode from any other modality's.

And there is no equivalent of pointing at the mistake: the channel is linear and has no spatial reference, so 'no, the other one' is genuinely hard to express and 'the third item' requires the system to have numbered them.

Which means a speech interface has to be designed around confirmation and correction rather than around input — and the confirmation cost is exactly what makes long speech interactions unpleasant, since every step needs one.

So speech suits short commands with clear confirmations, and suits dictation badly unless a visual channel is available for correction — **which is why dictation onto a screen works well and dictation to a speaker does not,** and the difference is the correction channel rather than the recognition quality.

3 Latency

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~100 ms    feels instantaneous. The action and the
           result are perceived as one event.

~1 s       the flow of thought is preserved, and the
           delay is noticed. No feedback needed beyond
           the result itself.

~10 s      attention is lost. The user will switch to
           something else, and needs a progress
           indicator and an estimate in order to
           decide whether to wait.

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AND THE SECOND-ORDER EFFECTS

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people type more slowly when the echo is delayed
they click again when nothing appeared to happen
they lose the causal connection between the
    action and the result, which breaks the gulf
    of evaluation (Module 01 section 1)
and VARIABLE latency is worse than uniformly slow
    latency, because it defeats prediction

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Variable latency is worse than uniformly slow latency — because the user cannot form an expectation, and so cannot plan around it: with a consistent two-second delay they learn to wait, and with a delay between 0.1 and four seconds they click again. **The variability finding is the one systems people need most**, because the usual optimisation target is the mean and **the tail is what determines the experience** (CSCE 670 Module 04 §4's same point about percentiles).

4 Choosing

- **What the hands and the eyes are doing. Driving, cooking, carrying something, and walking each remove a channel** — and **a design assuming full visual and manual attention fails in all four**, which covers a large fraction of mobile use.
- **The precision available. A finger is not a cursor, and a game controller is not either** — so the target sizes and in fact the whole layout change, rather than scaling (§1's nine millimetres).
- **The environment. Noise defeats speech; sunlight defeats low contrast; gloves defeat capacitive touch; cold defeats fine motor control** — **all of which are design constraints rather than user problems** (Module 01 §2).
- **And the social setting**, since **speech and gesture are public in a way that typing is not** — which is why voice assistants are used far less in offices and on public transport than their designers expected.
- **So the modality follows from the context**, and **a context nobody specified gets a desktop design** — **which is then deployed on a phone, in sunlight, to somebody holding a bag, and the failures are blamed on them**. Specifying the context is therefore part of the design rather than a preliminary to it.

AND THE ACCESSIBILITY CONNECTION

Every constraint in this module is somebody's permanent condition — a design that works one-handed, eyes-free, or without fine motor control works for far more people than the situational cases alone would suggest, because it works for both populations at once.

Which is CSCE 632's argument, and it is an argument about design quality rather than about accommodation — the constraint produces a better design for everybody, which is a stronger claim than fairness and does not depend on it.

And the situational framing is the persuasive one: 'usable while holding a child' and 'usable one-handed permanently' are precisely the same design requirement, and the first is easier to get approved.

So design for the constraint rather than for the population — which makes the requirement general and the benefit broad — and is where this module hands over to CSCE 632, which develops it properly.

Where to see this taught differently

Original material only goes so far — hearing the same idea in another voice is how it sticks. These are free, and they cover this module specifically.

Resource	What it covers for this module
MacKenzie — Fitts' law as a research and design tool (free)	§1 in depth — the formulations, the measurement method, and the two-dimensional extensions.
Card, Moran & Newell — The Psychology of Human-Computer Interaction	§1 and §3's foundations — where the timing constants in this module come from. Library copy.
Nielsen — Response Times: The 3 Important Limits (free)	§3's thresholds , with the sources and the behavioural consequences.
Material and Apple human interface guidelines (free)	§2's touch constraints as platform guidance — including the minimum target sizes and the reasons.

Exercises

Do these in code, not on paper, unless the exercise says otherwise. The module is not finished until they run.

- 1 **Measure your own Fitts's law constants** with a simple pointing task.
- 2 **Time hitting a corner against a same-sized target in the middle.**
- 3 **Find a destructive control adjacent to a common one** and move it.
- 4 **Measure a touch target in your own interface** and compare against nine millimetres.

- 5 **Find every hover-dependent element** in a desktop design and say what happens on touch.
- 6 **Try correcting a speech recognition error by voice alone**, and count the attempts.
- 7 **Add 300 ms of latency** to an interaction and observe your own behaviour.
- 8 **Add variable latency between 50 ms and 2 s** and compare.
- 9 **Use an interface one-handed** and list what fails.
- 10 **Specify the context** for something you are designing, in three sentences.

Self-check

Answer these in writing, from memory, without looking back at the notes. If you cannot, you have read this module but not yet learned it — that is useful information, not a failure.

- 1 State Fitts's law and what it predicts.
- 2 Why is an edge target effectively infinite, and why are corners fastest?
- 3 Give five design consequences, including the deliberate inverse use.
- 4 Name five input modalities with their error profiles.
- 5 What is the touch constraint most often forgotten?
- 6 Why does speech have no cheap undo, and what follows?
- 7 Give the three latency thresholds and what changes at each.
- 8 Why is variable latency worse than slow latency?
- 9 Name four things that determine the right modality.
- 10 State the accessibility connection and why it is a quality argument.